



**2004
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

Extension 2

Examiner: G. Rezcallah

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

☐ Mr Ee

○ Mr Rezcallah

Mr Barrett

Student Number:

(To be used by the exam markers only.)

[illegible]

QUESTION 1

Begin a new page.

	Marks
(a) Find: (i) $\int \frac{x^4}{\sqrt{x^5 - 7}} dx$.	2
(ii) $\int \frac{1}{e^x + e^{-x}} dx$	2
(b) Evaluate $\int_2^6 x\sqrt{6-x} dx$ Using the substitution $u^2 = 6 - x$.	3
(c) (i) Find the constants A and B such that $\frac{1}{\cos x} = \frac{A \cos x}{1 - \sin x} + \frac{B \cos x}{1 + \sin x}$	2
(ii) Hence, find the exact value of the integral $\int_0^{\frac{\pi}{6}} \sec x dx$	2
(d) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$	2
Hence, evaluate the integral $\int_0^{\frac{\pi}{2}} \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} dx$.	2

QUESTION 2

Begin a new page.

Marks

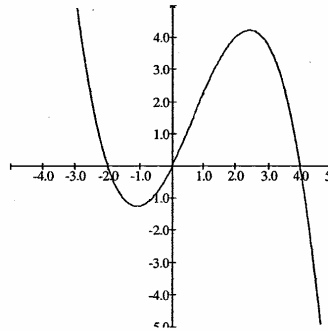
- (a) Consider the complex numbers $Z_1 = \sqrt{2}(1+i\sqrt{3})$ and $Z_2 = 2\sqrt{6}(1+i)$
- (i) Express $z = \frac{Z_1}{Z_2}$ exactly in the form $x + iy$, where x and y are real. 2
- (ii) Write Z_1 , Z_2 and z in modulus/argument form. 2
- (iii) Hence, find the exact value of $\cos \frac{\pi}{12}$. 1
- (iv) On an Argand diagram draw the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, and $\overrightarrow{OS}$, to represent Z_1 , Z_2 and $Z_1 - Z_2$ respectively. 2
- (b) Indicate on an Argand diagram the region which contains the point P representing z when :
- (i) $\operatorname{Re}(z + iz) \geq 2$ 2
- (ii) $1 \leq |z - 1 - i| \leq 3$ where $z = x + iy$ 2
- (c) By applying De Moivre's theorem and by also expanding $(\cos \theta + i \sin \theta)^5$, express $\sin 5\theta$ as a polynomial in $\sin \theta$. 4

QUESTION 3

Begin a new page.

Marks

- (a) The diagram shows the graph of $y = f(x)$ which passes through the origin and cuts the x axis at $x = -2$ and $x = 4$. The point $(1, 2\frac{1}{4})$ belongs to the curve.



- (i) Write down the equation of $y = f(x)$.

2

On separate diagrams, sketch each of the following:

10

(ii) $y = -f(x)$

(iii) $y = f(-x)$

(iv) $y^2 = f(x)$

(v) $y = |f(|x|)|$

(vi) $y = \frac{1}{1 - f(x)}$

- (b) Consider in the set of complex numbers $\mathbb{C}$:

w the cubic root of unity, $x = a + b$, $y = aw + bw^2$ and $z = aw^2 + bw$.

- (i) Show that $1 + w + w^2 = 0$

1

- (ii) Prove that $x^2 + y^2 + z^2 = 6ab$

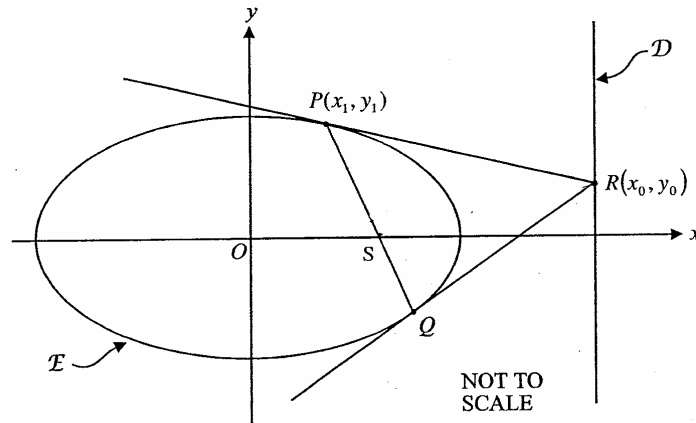
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QUESTION 4

Begin a new page.

Marks

(a)



The ellipse E with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ has a directrix D as shown in the diagram.

Point $R(x_0, y_0)$ lies on D .

PQ is the chord of contact from R where P is the point (x_1, y_1) .

- (i) Write down the equation of D and the coordinates of the focus S . 2
- (ii) Show that the tangent at a point $P(x_1, y_1)$ has an equation of $\frac{xx_1}{16} + \frac{yy_1}{9} = 1$. 2
- (iii) Write down the equation of chord PQ and show that the focus S lies on PQ . 2
- (iv) Show that the angle subtended by PR at the focus S is 90° . 3
- (v) Hence, deduce that the points P , S , and R are concyclic.. 1

QUESTION 4 (Continued)

Marks

(b)

(i) Sketch the graph of $f(x) = \sqrt{(x+1)^2} + \sqrt{(x-1)^2}$. 2

(ii) Hence, solve the equation $-2 \leq \sqrt{(x+1)^2} + \sqrt{(x-1)^2} < 2$ 1

(c) The equation $x^3 + kx + r = 0$ has roots α , β , and γ .

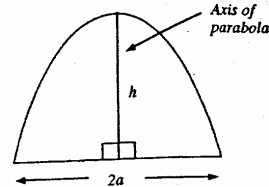
Find the value of the expression $\alpha^3 + \beta^3 + \gamma^3$ 2

QUESTION 5 Begin a new page.

Marks

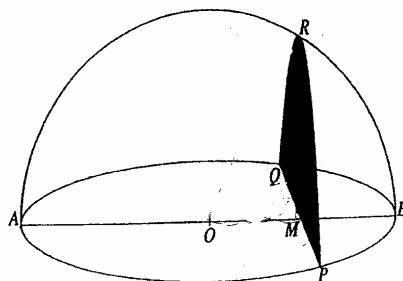
- (a) (i) A parabolic segment has height h and width $2a$.

Use Simpson's rule with three function values to show that the exact area of this segment is $\frac{4}{3}ah$.



2

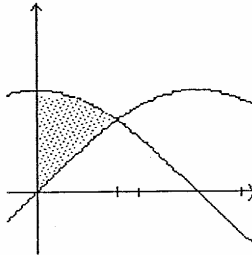
In the diagram below, a tent has a circular base with centre O and radius a , and AOB is a diameter of the base. - The shaded area $PMQR$ is a typical cross section of the tent perpendicular to AB , and meets AB at a point M distant x from O . The curve PRQ is a parabola with axis RM and $QM = RM$.



- (ii) Use part (i) to show that the shaded area $PMQR$ is $\frac{4}{3}(a^2 - x^2)$. 2
- (iii) Find the volume of the tent. 2
- (b) Factorise the polynomial $P(z) = z^4 - 2z^2 + 8z - 3$ fully over $\mathbb{C}$. 4
Given that $P(1 - \sqrt{2}i) = 0$.
- (c) (i) By considering the perfect square $(\sqrt{x} - \sqrt{y})^2$ where x and y are positive, prove that $\frac{x+y}{2} \geq \sqrt{xy}$. 2
- (ii) Hence, if a, b, c and d are positive numbers, prove that : 3
- $$4(ab + bc + cd + da) \leq (a + b + c + d)^2$$

QUESTION 6 Begin a new page.**Marks**

- (a) In the diagram, the shaded region is bounded by the y axis and the curves $y = \cos x$ and $y = \sin x$.



- (i) Show that the curves intersect at $\left(\frac{\pi}{4}, \frac{1}{\sqrt{2}}\right)$ 1
- (ii) The shaded region is rotated about the y axis.
Find the exact value of the volume obtained by this rotation, using the method of cylindrical shells. 5
- (b) By considering the binomial expansion of $(1+i)^n$
Show that $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots = 2^{\frac{n}{2}} \cos \frac{n\pi}{4}$ 2
- (c) The normal at a point $P\left(x_1, \frac{9}{x_1}\right)$ on the rectangular hyperbola $xy = 9$ meets the curve again at another point A.
- (i) Prove that the equation of intersection of the normal at P and the rectangular hyperbola is $x_1^3 x^2 + (81 - x_1^4)x - 81x_1 = 0$ 2
- (ii) Hence, prove that the coordinates of A are $\left(-\frac{81}{x_1^3}, -\frac{x_1^3}{9}\right)$ 2
- (iii) Let M be the midpoint of AP. Derive the cartesian equation of the locus of M. 3

QUESTION 7 Begin a new page.**Marks**

- (a) Consider the word
- SOCCER**
- .

How many:

- (i) six-letter different arrangements 1
- (ii) selections of 4 letters 2

can be made from the letters in the word **SOCCER**?

- (b) A particle of mass
- m
- is projected vertically upward under gravity in a medium in which the resistance is proportional to square of the velocity (
- mkv^2
-), where
- k
- is a constant.

- (i) Show that the terminal speed
- V
- in the medium is
- $\sqrt{\frac{g}{k}}$
- 1

If the speed of projection is equal to the terminal velocity V in the medium, show that:

- (ii) the particle reaches a maximum height of
- $\frac{V^2}{2g} \ln 2$
- above the point of projection. 3

- (iii) the time taken to reach its maximum height is
- $\frac{\pi V}{4g}$
- 4

- (iv) the time
- t
- in the downward motion as a function of
- v
- is given by

$$t = \frac{1}{2\sqrt{gk}} \ln \left(\frac{V+v}{V-v} \right) \quad 4$$

QUESTION 8 Begin a new page.

Marks

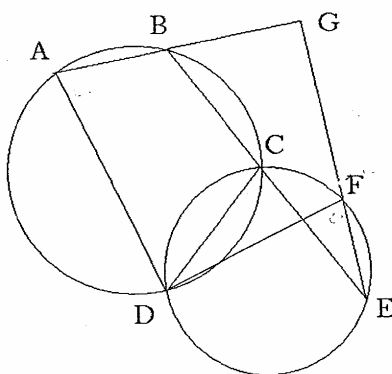
(a) If $I_n = \int_0^1 (1-x^2)^n dx$ for $n \geq 0$, show that $I_n = \frac{2n}{1+2n} I_{n-1}$ for $n \geq 1$.

3

Hence, find an expression for I_n in terms of n for $n \geq 1$.

2

(b)



Two circles intersect at C and D. ABCD is a cyclic quadrilateral in one circle.

BC produced meets the other circle at E. C, F, E and D are concyclic points.

AB produced meets EF produced at G.

Prove that GFDA is a cyclic quadrilateral.

4

(c) A sequence is defined by the recurrence relationship:

$$U_1 = 1 \text{ and } U_{n+1} = \frac{1}{2} \left[U_n + \frac{2}{U_n} \right] \text{ when } n \geq 1, n \text{ a positive integer}$$

(i) Prove by mathematical induction: $\frac{U_n - \sqrt{2}}{U_n + \sqrt{2}} = \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^{2^{n-1}}$

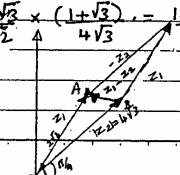
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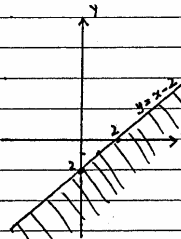
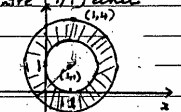
(ii) Hence, show that for n sufficiently large, U_n is very close to $\sqrt{2}$

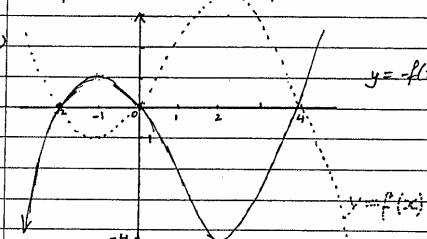
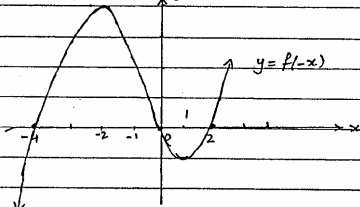
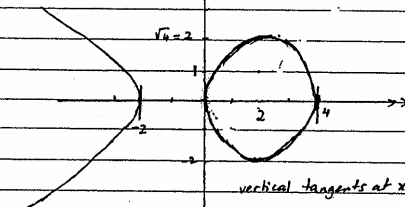
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Solution of NSBHS Ext. II - 2004 HSC Trial.	
Question 1.	Suggested Marks.
(a) (i) $\int \frac{x^4 dx}{\sqrt{x^5-7}}$ let $u = x^5-7$ $du = 5x^4 dx$	1 for correct substitution
$\frac{1}{5} \int \frac{5x^4 dx}{\sqrt{x^5-7}} = \frac{1}{5} \int \frac{du}{\sqrt{u}} = \frac{1}{5} \int u^{-\frac{1}{2}} du$	2 for correct modified primitive
$= \frac{1}{5} 2u^{\frac{1}{2}} + C = \frac{2}{5} \sqrt{x^5-7} + C$	1 for correct answer w/c
(ii) $\int \frac{dx}{e^x + e^{-x}} = \int \frac{dx}{e^x + \frac{1}{e^x}} = \int \frac{e^x dx}{e^{2x} + 1}$	1 for correct start and substitution
let $u = e^x$ $du = e^x dx$	1 for correct answer in terms of x.
$= \int \frac{du}{u^2 + 1} = \tan^{-1} u + C = \tan^{-1} e^x + C$	
(b) $\int_2^6 x \sqrt{6-x} dx$ $u^2 = 6-x \Rightarrow x = 6-u^2$ $2u du = -dx \Rightarrow dx = -2u du$	
$\int (6-u^2) \sqrt{u^2} \cdot (-2u du)$	1 for substitution to obtain $\int$ in terms of u.
$= \int_2^0 (6-u^2)(-2u^2) du = -2 \int_2^0 (6u^2 - u^4) du$	
$= -2 \left[\frac{6u^3}{3} - \frac{u^5}{5} \right]_2^0 = 2 \left[2u^3 - \frac{u^5}{5} \right]_0^2$	1 for finding primitive with correct bounds.
$= 2 \left[2(2)^3 - \frac{2^5}{5} \right] = 2 \left[16 - \frac{32}{5} \right] = \frac{96}{5}$	1 for correct answer.
(c) (i) $\frac{1}{\cos x} = \frac{A \cos x (1 + \sin x) + B \cos x (1 - \sin x)}{1 - \sin^2 x}$	Award:
$= \frac{A \cancel{\cos x} (1 + \sin x) + B \cancel{\cos x} (1 - \sin x)}{\cancel{\cos^2 x}}$	1 mark for one correct answer or correct 2 equations in terms of A and B.
$= \frac{A(1 + \sin x) + B(1 - \sin x)}{\cos x}$	
$= \frac{(A+B) + (A-B) \sin x}{\cos x}$	2 marks for both correct answers.
$A+B=1$ $A-B=0 \Rightarrow A=B$	$\Rightarrow 2A=1 \Rightarrow A=\frac{1}{2}$ $B=\frac{1}{2}$ ✓

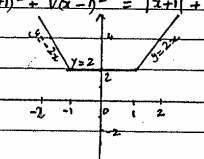
NSBHS - Ext II Trial 2004 Solutions.	Marks.
$(ii) \int_0^{\pi/6} \sec x \, dx = \int_0^{\pi/6} \frac{1}{\cos x} \, dx = \frac{1}{2} \int_0^{\pi/6} \frac{\cos x \, dx}{1 - \sin x} + \frac{1}{2} \int_0^{\pi/6} \frac{\cos x \, dx}{1 + \sin x}$ $= -\frac{1}{2} \int_0^{\pi/6} \frac{\cos x \, dx}{1 - \sin x} + \frac{1}{2} \int_0^{\pi/6} \frac{\cos x \, dx}{1 + \sin x}$ $= -\frac{1}{2} \left[\ln(1 - \sin x) \right]_0^{\pi/6} + \frac{1}{2} \left[\ln(1 + \sin x) \right]_0^{\pi/6}$ $= \frac{1}{2} \left[\ln \frac{(1 + \sin x)}{(1 - \sin x)} \right]_0^{\pi/6} = \frac{1}{2} \ln \left(\frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} \right)$ $= \frac{1}{2} \ln \frac{3}{1} = \frac{1}{2} \ln 3 = \ln \sqrt{3}$	1 for correct substitution to get the correct primitive in either 1 mark for correct answer.
$(d) (i) \int_a^b f(x) \, dx = \int_a^b f(a-x) \, dx$ $\text{let } u = a-x \Rightarrow \int_a^b f(a-x) \, dx = \int_a^0 f(u) \, du$ $du = -dx$ $= \int_0^a f(u) \, du = \int_0^a f(x) \, dx$ $\text{or: } \int_a^b f(x) \, dx = - \int_a^0 f(a-u) \, du = \int_0^a f(a-u) \, du$ $= \int_0^a f(a-x) \, dx$ $I = \int_0^{\pi/2} \frac{\sin x}{e^{\sin x} + e^{\cos x}} \, dx = \int_0^{\pi/2} \frac{e^{\sin(\frac{\pi}{2}-x)}}{e^{\sin(\frac{\pi}{2}-x)} + e^{\cos(\frac{\pi}{2}-x)}} \, dx$ $= \int_0^{\pi/2} \frac{e^{\cos x}}{e^{\cos x} + e^{\sin x}} \, dx$ $I + I = \int_0^{\pi/2} \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} \, dx + \int_0^{\pi/2} \frac{e^{\cos x}}{e^{\sin x} + e^{\cos x}} \, dx$ $= \int_0^{\pi/2} \frac{(e^{\sin x} + e^{\cos x})}{(e^{\sin x} + e^{\cos x})} \, dx = \int_0^{\pi/2} 1 \, dx = \left[x \right]_0^{\pi/2}$ $\therefore 2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$	Award: 2 marks for correct proof. 1 mark for any minor step not shown in the proof. 1 mark for correct method using property. 1 for correct answer.

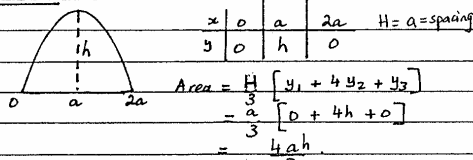
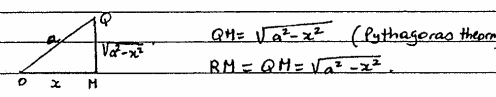
NSBHS Trial Ext 2 Solutions.	Marks.
Question 2.	
$(a) Z_1 = \sqrt{2}(1+i\sqrt{3}) \quad Z_2 = 2\sqrt{6}(1+i)$	Award:
$(i) z = \frac{Z_1}{Z_2} = \frac{\sqrt{2}(1+i\sqrt{3})}{2\sqrt{6}(1+i)} \times \frac{(1-i)}{(1-i)}$ $= \frac{\sqrt{2}(1-i-i\sqrt{3}-i^2\sqrt{3})}{2\sqrt{6}(1-i^2)}$ $= \frac{\sqrt{2}(1-i-i\sqrt{3}+\sqrt{3})}{2\sqrt{6}(2)}$ $= \frac{(1+\sqrt{3}-i-i\sqrt{3})}{4\sqrt{3}}$ $= \frac{1+\sqrt{3}}{4\sqrt{3}} + i \frac{(\sqrt{3}-1)}{4\sqrt{3}}$	2 marks for correct z answer. 1 mark for : correct method, wrong answer or 1 correct part of z showing (1-i)
$(ii) Z_1 = \sqrt{2}(1+i\sqrt{3}) \quad r = \sqrt{1^2 + (\sqrt{3})^2} = 2$ $= \frac{2\sqrt{2}(1+i\sqrt{3})}{2} = 2\sqrt{2} \operatorname{cis} \frac{\pi}{3}$ $Z_2 = 2\sqrt{6}(1+i) = 2\sqrt{6} \times \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) \times \sqrt{2}$ $= 4\sqrt{3} \left(\frac{1}{2} + i \frac{1}{2} \right) = 4\sqrt{3} \operatorname{cis} \frac{\pi}{4}$ $z = \frac{Z_1}{Z_2} = \frac{2\sqrt{2} \operatorname{cis} \frac{\pi}{3}}{4\sqrt{3} \operatorname{cis} \frac{\pi}{4}} = \frac{\sqrt{2}}{2\sqrt{3}} \operatorname{cis} \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$ $= \frac{\sqrt{2}}{2\sqrt{3}} \operatorname{cis} \frac{\pi}{12}$	1 mark for correct z according to z₁ and z₂.
$(iii) z = \frac{\sqrt{2}}{2\sqrt{3}} \operatorname{cis} \frac{\pi}{12} = \frac{\sqrt{2}}{2\sqrt{3}} \left[\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right]$ $\therefore \left[\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right] = \frac{2\sqrt{3}}{\sqrt{2}} z = 2\sqrt{3} \left[\frac{1+\sqrt{3}}{4\sqrt{3}} + i \frac{(\sqrt{3}-1)}{4\sqrt{3}} \right]$ $\therefore \cos \frac{\pi}{12} = \frac{2\sqrt{3}}{\sqrt{2}} \times \frac{(1+\sqrt{3})}{4\sqrt{3}} = \frac{1+\sqrt{3}}{2\sqrt{2}}$	1 mark for the correct expression (even if unsimplified).
(iv) 	1 mark for correct representation of $\vec{OA}$ and $\vec{OB}$ 1 mark for $\vec{BA}$.

NSBHS Ext 2 Solutions.	Harks.
(b) (i) $\operatorname{Re}(z + iz) \geq 2$. $z + iz = x + iy + i(x + iy)$ $= x + iy + ix - y$ $= (x - y) + i(x + y)$ $\operatorname{Re}(z + iz) = x - y \geq 2$ or $y \leq x - 2$. 	1 mark for the correct line 1 mark for the correct region.
(ii) $z - (1 + i) = 3$ is a circle of centre $(1, 1)$ and $r = 3$. $z - (1 + i) = 1$ is a circle of centre $(1, 1)$ and radius = 1. 	1 mark for identifying and showing the 2 circles with their correct centres and radii 1 mark for the shaded region.
(c) $(\cos \theta + i \sin \theta)^5$. Let $c = \cos \theta$, $s = \sin \theta$. $(c + is)^5$ $= c^5 + 5c^4si + 10c^3s^2 + 10c^2(is)^3 + 5c(is)^4 + (is)^5$ $= c^5 + 5c^4si - 10c^3s^2 - 10c^2s^3 + 5cs^4 - is^5$ $= c^5 - 10c^3s^2 + 5cs^4 + i(5c^4s - 10c^2s^3 + s^5)$ 1 mark for correct expansion. But $(\cos \theta + i \sin \theta)^5 = \cos 5\theta + i \sin 5\theta$. (De Moivre's theorem) $\therefore \sin 5\theta = 5c^4s - 10c^2s^3 + s^5$ $= 5c^4s - 10(1 - s^2)s^3 + s^5$ $= 5(1 - s^2)^4s - 10s^3 + s^5$ $= 5s^5 - 10s^3 + 5s - 10s^3 + 11s^5$ $= 16s^5 - 10s^3 + 5s$ $= 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$ 1 mark for correct answer. $\therefore \sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$.	

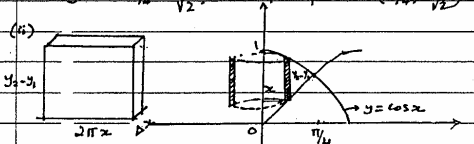
NSBHS Ext 2 - Solutions 2004	Harks.
Question 3 (a) (i) $y = ax(x - 4)(x + 2)$ $2\frac{1}{4} = a(1 - 4)(1 + 2)$ $\frac{9}{4} = -9a \Rightarrow a = -\frac{1}{4}$ $y = -\frac{1}{4}x(x - 4)(x + 2)$ or $-\frac{1}{4}(x^3 - 2x^2 - 8x)$ (ii)  (iii)  (iv) 	1 mark for quadratic showing factors. 1 for correct expression, either factored or simplified form. correct graph: ✓ Award: Correct graph ✓✓ substantially correct, but missing feature ✓ Award: 1 mark for either: correct loop or correct ✓ or correct $y = \sqrt{f(x)}$. 2 marks for correct graph, including vertical tangents.

NSBHS Extension 2 - 2004 Total solution	Marks
Question 4.	
(i) $b^2 = a^2(1 - e^2)$ $\frac{x^2}{16} + \frac{y^2}{9} = 1$	
$9 = 16(1 - e^2) \Rightarrow 9 = 16 - 16e^2 \Rightarrow 16e^2 = 16 - 9 = 7$	
$e = \frac{\sqrt{7}}{4}$	
Directrix $x = \frac{a}{e} = \frac{4}{\frac{\sqrt{7}}{4}} = \frac{16}{\sqrt{7}}$ or $\frac{16\sqrt{7}}{7}$	✓ for correct directrix
$S(ae, 0) = S(\frac{4\sqrt{7}}{4}, 0) = S(\sqrt{7}, 0)$	✓ for correct focus
(ii) $\frac{x^2}{16} + \frac{y^2}{9} = 1$	
$\frac{2x}{16} + \frac{2yy'}{9} = 0 \Rightarrow \frac{xy'}{9} = -\frac{x}{16}$	
$y' = -\frac{9x}{16y} \Rightarrow m = -\frac{9x_1}{16y_1}$	✓ for correct m
$y = y_1 - m(x - x_1)$	
$y = y_1 + \frac{9x_1}{16y_1}(x - x_1)$	
$16yy_1 - 16y_1^2 = -9x_1x + 9x_1^2$	
$9x_1x + 16yy_1 = 9x_1^2 + 16y_1^2 \div 9 \times 16$	✓ for the correct manipulation of the one-pt formula
$\frac{x_1x}{16} + \frac{yy_1}{9} = \frac{x_1^2}{16} + \frac{y_1^2}{9}$ But $\frac{x_1^2}{16} + \frac{y_1^2}{9} = 1$	
$\frac{x_1x}{16} + \frac{yy_1}{9} = 1$	
(iii) $\frac{x_0x}{16} + \frac{y_0y}{9} = 1$ is the chord PQ equation.	✓ for PQ equation in general
R(x ₀ , y ₀) lies on D $\therefore x_0 = \frac{16}{\sqrt{7}}$	
$\frac{16}{\sqrt{7} \times 16}x + \frac{y_0y}{9} = 1$	
$\frac{x}{\sqrt{7}} + \frac{yy_0}{9} = 1$ is equation of PQ.	✓ for showing S ∈ PQ
Sub $S(\sqrt{7}, 0)$ into $\frac{x}{\sqrt{7}} + \frac{yy_0}{9} = 1 \Rightarrow 1 = 1 - \text{RHS} \therefore S \in PQ$	
(iv) $\frac{x_1x}{16} + \frac{yy_1}{9} = 1$ is the tangent.	
At $x = \frac{16}{\sqrt{7}}$; $\frac{16}{\sqrt{7}} \times \frac{x_1}{16} + \frac{yy_1}{9} = 1 \Rightarrow \frac{y_1y}{9} = 1 - \frac{x_1}{\sqrt{7}}$	
$y = \frac{9}{y_1} \left(1 - \frac{x_1}{\sqrt{7}}\right) = \frac{9}{\sqrt{7}y_1}(\sqrt{7} - x_1)$	✓ for finding y _k in terms of y ₁
$R\left(\frac{16}{\sqrt{7}}, \frac{9}{\sqrt{7}y_1}(\sqrt{7} - x_1)\right)$	

NSBHS Extension 2 Total HSC - 2004	Marks
$m_{SR} = \frac{9(\sqrt{7} - x_1)}{16y_1} = \frac{9(\sqrt{7} - x_1)}{\frac{y_1 - y_1}{16 - 7}} = \frac{9(\sqrt{7} - x_1)}{\frac{y_1}{9}}$	✓ for gradient SR or PS
$m_{SR} = \frac{(\sqrt{7} - x_1)}{y_1}$	
$m_{PS} = \frac{y_1 - 0}{x_1 - \sqrt{7}} = \frac{y_1}{x_1 - \sqrt{7}}$	
$m_{PS} \times m_{SR} = \frac{(\sqrt{7} - x_1)}{y_1} \times \frac{y_1}{(x_1 - \sqrt{7})} = -\frac{(x_1 - \sqrt{7})}{(x_1 - \sqrt{7})} = -1$	✓ for proving the product of gradients is -1
$\therefore PS \perp SR \Rightarrow \angle PSR = 90^\circ$	
(v) P, S, R are concyclic as the $\angle$ in a semi-circle is 90° (with PR as diameter).	
(b) (i) $f(x) = \sqrt{(x+1)^2} + \sqrt{(x-1)^2} = x+1 + x-1 $	
	✓ Award: 2 marks for correct graph
(ii) Impossible no real solution.	✓ 0 for non linear curve. 1 mark for part of the graph correct showing intercepts or for the correct equations. 1 for correct (ii) answer according to graph
(c) $x^3 + kx + r = 0$	
$x^3 + kx + r = 0 \quad \alpha + \beta + \gamma = -\frac{b}{a} = 0$	
$x^3 + kx + r = 0$	
$x^3 + kx + r = 0$	
$x^3 + \beta^3 + \gamma^3 + k(\alpha + \beta + \gamma) + 3r = 0$	✓ 1 for correct method
$x^3 + \beta^3 + \gamma^3 = -3r - k(\alpha + \beta + \gamma)$	
$= -3r$	✓ 1 for correct answer

NSBHS Extension 2 Trial HSC. - 2004		Marks
Question 5:		
(a) (i)		✓ for correct setting ✓ for correct application of method leading to given answer.
(ii)		✓ for QM or RM
(iii)	$V = \lim_{a \rightarrow \infty} \int_{x=-a}^x \Delta x$ $= \int_{-a}^a \frac{4}{3} (a^2 - x^2) dx = 2 \int_0^a \frac{4}{3} (a^2 - x^2) dx$ $= \frac{8}{3} \left[a^2 x - \frac{x^3}{3} \right]_0^a = \frac{8}{3} \left[a^3 - \frac{a^3}{3} \right] = \frac{8}{3} \times \frac{2a^3}{3} = \frac{16}{9} a^3$	✓ for correct method using (i) to get the given answer. ✓ for correct integral ✓ for correct answer
(b)	$z^4 - 2z^2 + 8z - 3 = 0$ $P(1 - \sqrt{2}i) = 0 \Rightarrow$ the conjugate is a root $\Rightarrow P(1 + \sqrt{2}i) = 0$ $[z - (1 - \sqrt{2}i)][z - (1 + \sqrt{2}i)] = 0$ ✓ Sum = $1 - \sqrt{2}i + 1 + \sqrt{2}i = 2$ Product = $(1 - \sqrt{2}i)(1 + \sqrt{2}i) = 1 - 2i^2 = 3$ $\Rightarrow z^2 - Sz + P = 0 \Rightarrow z^2 - 2z + 3 = 0$ is factor. $\begin{array}{r} z^2 + 2z - 1 \\ z^4 - 2z^2 + 8z - 3 \\ \hline 2z^3 - 5z^2 + 8z - 3 \\ \ominus 2z^3 + 4z^2 + 6z \\ \hline -z^2 + 2z - 3 \\ \oplus z^2 - 2z + 3 \\ \hline 0 \end{array}$	✓ for conjugate factor ✓ for correct answer of division

NSBHS Extension 2 Trial Solutions - 2004		Marks
Solving: $z^2 + 2z - 1 = 0$		
$z = \frac{-2 \pm \sqrt{4 - 4(-1)}}{2} = \frac{-2 \pm \sqrt{8}}{2} = \frac{-2 \pm 2\sqrt{2}}{2}$ $z = -1 \pm \sqrt{2}$		✓ for correct real factors.
$\therefore P(z) = (z - (1 - \sqrt{2}i))(z - (1 + \sqrt{2}i))(z - (-1 + \sqrt{2}))$ $= (z - (1 - \sqrt{2}i))(z - (-1 - \sqrt{2}))$ $= (z - 1 + \sqrt{2}i)(z - 1 - \sqrt{2}i)(z + 1 + \sqrt{2})$		✓ for factors over C
(c) (i) $(\sqrt{x} - \sqrt{y})^2 \geq 0$		
$x + y - 2\sqrt{xy} \geq 0$ $x + y \geq 2\sqrt{xy}$ $\frac{x+y}{2} \geq \sqrt{xy}$		✓ for correct start.
(ii) let $x = a + b$ $y = c + d$		✓ for correct method.
From (i) $\frac{x+y}{2} \geq \sqrt{xy}$ $\frac{a+b+c+d}{2} \geq \sqrt{(a+b)(c+d)}$ $\frac{a+b+c+d}{4} \geq \sqrt{\frac{ac+ad+bc+bd}{4}}$		✓ for applying (i)
Square both sides: $(\frac{a+b+c+d}{2})^2 \geq \frac{ac+ad+bc+bd}{1}$ $\times 16$ $(a+b+c+d)^2 \geq 4(ac+ad+bc+bd)$ $\therefore 4(ac+ad+bc+bd) \leq (a+b+c+d)^2$		✓ for manipulation
N.B: some students' solutions may be: $a+b \geq 2\sqrt{ab}$ $b+c \geq 2\sqrt{bc}$ $+ c+d \geq 2\sqrt{cd}$ $a+d \geq 2\sqrt{ad}$ $2(a+b+c+d) \geq 2(\sqrt{ab} + \sqrt{bc} + \sqrt{cd} + \sqrt{ad})$ $(a+b+c+d) \geq (\sqrt{ab} + \sqrt{bc} + \sqrt{cd} + \sqrt{ad})$		✓ for getting the result.
		← Award 1 mark for this.

NSBHS Extension 2 - Maths Trial 2004.	Marks.
Question 6.	
(a) (i) $y = \sin x = \cos x$ at: sub $x = \pi/4$ and $y = \frac{1}{\sqrt{2}}$ in both.	
$\sin x = 1$ $\cos x = 1$ $\tan x = 1$ $\therefore x = \pi/4$.	✓ for showing.
$y = \sin \pi/4 = \frac{1}{\sqrt{2}}$. $\therefore$ pt of $\cap$ is $(\pi/4, \frac{1}{\sqrt{2}})$	
(ii) 	
$A_x = 2\pi x (y_2 - y_1)$ $= 2\pi x (\cos x - \sin x)$	✓ for correct area.
$V_x = 2\pi x (\cos x - \sin x) \Delta x$ $V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/4} 2\pi x (\cos x - \sin x) \Delta x$ $= \int_0^{\pi/4} 2\pi x (\cos x - \sin x) dx$ $= 2\pi \int_0^{\pi/4} x (\cos x - \sin x) dx$	✓ for correct V. expression.
Let $u = x$ $du = dx$ $v = \cos x - \sin x$ $dv = (-\sin x + \cos x) dx$	✓ for correct method start of integration by parts
$= uv - \int v du$ $= x(\sin x + \cos x) - \int (\cos x - \sin x) dx$ $= x(\sin x + \cos x) - [-\sin x + \cos x]$ $= x(\sin x + \cos x) + \sin x - \cos x$ $V = 2\pi \left[\left[x(\sin x + \cos x) + \sin x - \cos x \right]_0^{\pi/4} \right]$ $= 2\pi \left\{ \frac{\pi}{4} \left[\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right] + \left[\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - (1) \right] \right\}$ $= 2\pi \left[\frac{\pi}{4} \times \sqrt{2} - 1 \right] = 2\pi \left[\frac{\sqrt{2}\pi - 4}{4} \right]$ $= \pi \left[\frac{\sqrt{2}\pi - 4}{2} \right]$ cubic unit. Accept: $\frac{2\pi^2}{4} \left(\frac{\sqrt{2}}{2} - 1 \right)$	✓ for correct integration process. ✓ for correct answer.

NSBHS Extension 2 - Maths Trial 2004	Marks.
2 nd Approach to Q6 (a) (ii)	Marks to Q6 a (ii) again.
(ii) After obtaining: $V = 2\pi \int_0^{\pi/4} x (\cos x - \sin x) dx$ $= 2\pi \left[\int_0^{\pi/4} x \cos x dx - \int_0^{\pi/4} x \sin x dx \right]$	1 mark for correct A 1 mark for correct V.
$I_1 = \int_0^{\pi/4} x \cos x dx$ $= \left[x \sin x \right]_0^{\pi/4} - \int_0^{\pi/4} \sin x dx$ $= \left[x \sin x + \cos x \right]_0^{\pi/4} = \frac{\pi}{4} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1$	1 mark for correct Integration by parts
$I_2 = \int_0^{\pi/4} x \sin x dx$ $= \left[-x \cos x \right]_0^{\pi/4} + \int_0^{\pi/4} \cos x dx$ $= \left[-x \cos x + \sin x \right]_0^{\pi/4} = -\frac{\pi}{4} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$	1 mark for correct process.
$V = 2\pi [I_1 - I_2]$ $= 2\pi \left[\frac{\pi}{4} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 + \frac{\pi}{4} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right]$ $= 2\pi \left[\frac{2\pi}{4\sqrt{2}} - 1 \right] = \frac{2\pi^2}{2\sqrt{2}} - 2\pi$ $= \frac{\pi^2}{\sqrt{2}} - 2\pi$	1 mark for correct answer.
3 rd approach to $\int$:	or
(ii) to get the volume $\int$ in one line: $2\pi \int_0^{\pi/4} (x \sin x + \cos x) + (x \cos x - \sin x) dx$ $= 2\pi \left[\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} - 1 + \frac{\pi}{4\sqrt{2}} - \frac{1}{\sqrt{2}} \right]$ $= 2\pi \left[\frac{2\pi}{4\sqrt{2}} - 1 \right] = \frac{\pi^2}{\sqrt{2}} - 2\pi$	✓ ✓
Note: Some students start wrong to get: $2\pi \int_0^{\pi/4} (\cos x - \sin x) dx$ and get the correct answer to that which is $\frac{\pi^2}{4}$	→ to get $\pi^2/4$, students will get 3 marks.

NSBHS Extension 2 - Trial 2004.	Marks.
(b) $(1+i)^n = [\sqrt{2} \text{cis } \pi/4]^n$ De Moivre's thm. Expanding $(1+i)^n$ $= 1 + \binom{n}{1}i + \binom{n}{2}i^2 + \binom{n}{3}i^3 + \binom{n}{4}i^4 + \binom{n}{5}i^5 + \binom{n}{6}i^6 + \dots$ $= 1 + \binom{n}{1}i - \binom{n}{2} + \binom{n}{3}i - \binom{n}{4} + \binom{n}{5}i - \binom{n}{6} + \dots$ (knowing $i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i, i^6 = -1, \dots$) $= 1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots + i \left[\binom{n}{1} - \binom{n}{3} + \binom{n}{5} - \binom{n}{7} + \dots \right]$ ✓ for expansion. Put $(1+i)^n = (\sqrt{2})^n \left(\cos n\pi/4 + i \sin n\pi/4 \right)$ as $\sqrt{2} = 2^{1/2}$ $= 2^{n/2} \cos n\pi/4 + i 2^{n/2} \sin n\pi/4$ ✓ for De Moivre's expansion. Equating real sides: $1 - \binom{n}{2} + \binom{n}{4} - \binom{n}{6} + \dots = 2^{n/2} \cos n\pi/4$ (c) i) $xy = 9 \Rightarrow y = \frac{9}{x} = 9x^{-1}$ $y' = -9x^{-2} = -\frac{9}{x^2}$ tangent $m_1 = -\frac{9}{x_1^2} \Rightarrow$ normal $m = \frac{x_1^2}{9}$ $y - y_1 = \frac{x_1^2}{9} (x - x_1)$ $y = \frac{x_1^2}{9} x - \frac{x_1^3}{9} + y_1$ is normal. It intersects hyperbola: $y = \frac{9}{x}$ $\frac{x_1^2}{9} x - \frac{x_1^3}{9} + y_1 = \frac{9}{x}$ but $y_1 = \frac{9}{x_1}$ $\frac{x_1^2}{9} x - \frac{x_1^3}{9} + \frac{9}{x_1} = \frac{9}{x}$ $\times 9x = x_1$ $x_1^3 x^2 - x_1^4 x + 81x = 81x_1$ $x_1^3 x^2 + (81 - x_1^4)x - 81x_1 = 0$ ✓ for equating + manipulating to get the given relation. (ii) quadratic equations whose roots should be x of P and A. i.e. $x_P = x_1$ and $x_A = \alpha$. Use Sum = $-\frac{b}{a}$. $\odot$ $P = -81x_1$ $x_1 + \alpha = -\frac{81 + x_1^4}{x_1^3}$ $\alpha \cdot x_1 = -\frac{81x_1}{x_1^3}$ $x_1/\alpha = -\frac{81}{x_1^3} + \frac{1}{x_1}$ $\alpha = -\frac{81}{x_1^3}$ $y_A = +\frac{9}{x_A} = +\frac{9}{-\frac{81}{x_1^3}} = -\frac{x_1^3}{9}$ ✓ for showing x_A. [Accept substitution if fully shown] (OR) $(x - x_1)(x^2 x + 81) = 0$ (Factoring)	

NSBHS Ext. 2 Maths Trial - 2004.	Marks.
(iii) A $\left(-\frac{81}{x_1^3}, -\frac{x_1^3}{9} \right)$ P $\left(x_1, \frac{9}{x_1} \right)$ $X = X_H = \frac{x_A + x_P}{2} = \frac{-\frac{81}{x_1^3} + x_1}{2} = \frac{-81 + x_1^4}{2x_1^3} \dots (1)$ $Y = \frac{y_A + y_P}{2} = \frac{-\frac{x_1^3}{9} + \frac{9}{x_1}}{2} = \frac{-x_1^4 + 81}{9x_1} \dots (2)$ ✓ for both coordinates of A. (2) $Y = \frac{81 - x_1^4}{18x_1}$ (1) $X = \frac{81 - x_1^4}{18x_1} \div \frac{x_1^4 - 81}{2x_1^3} = -\frac{x_1^2}{9}$ $\Rightarrow x_1^2 = -\frac{9Y}{X} \dots (3)$ Using (2) again: $Y = \frac{81 - x_1^4}{18x_1} \dots (3)$ Square both sides: $Y^2 = \frac{(81 - x_1^4)^2}{18^2 x_1^2}$ but $x_1^2 = -\frac{9Y}{X} \Rightarrow Y^2 = \frac{(81 - (-\frac{9Y}{X})^2)^2}{18^2 (-\frac{9Y}{X})}$ ✓ trying to get rid of x_1 in $x_1^2 = -\frac{9Y}{X}$ $18^2 \left(-\frac{9Y}{X} \right) Y^2 = \left(81 - \frac{81Y^2}{X^2} \right)^2$ $-18^2 (9) Y^3 = \frac{(81x^2 - 81Y^2)^2}{X^4} \times X^4$ $\div 9^3 = 729$ ✓ 1 for getting the relation. $-4Y^3 X^3 = 9(X^2 - Y^2)^2$ $\therefore 4X^3 Y^3 = -9(X^2 - Y^2)^2$ OR $4X^3 Y^3 = 9(Y^2 - X^2)^2$	

NSBHS Extension 2 - Maths Trial 2004.		Marks
<u>Question 7:</u>		
(a) SDCCER		
(i) 2 C's : $\frac{6!}{2} = 360$		✓ for 6! or answer
(ii) 2C's 4C_2 4 different letters		
1C, 3 others 4C_1 or: from S, D, C, E, R: 5C_4		✓ for correct method
OC, 4 others 4C_0 + 2C's, 2 diff: 4C_2		
${}^4C_2 + {}^4C_1 + {}^4C_0 = 11$ ${}^5C_4 + {}^4C_2 = 5+6=11$		✓ for answer.
NB: If students wrote: ${}^4C_2 + {}^2C_2 + {}^2C_1 \times {}^4C_3 + {}^2C_0 \times {}^4C_4 = 15 \rightarrow$ they get only 1 mark.		
(b) (i) Downward Motion is the only motion to get the terminal velocity:		
$\uparrow mkv^2$ $m\ddot{x} = mg - mkv^2$		✓ 1 mark to show the correct downward
$\downarrow mg$ $\ddot{x} = g - kv^2$		$\ddot{x} = 0$.
Terminal velocity: $\ddot{x} = 0$		
$g - kv^2 = 0$		[No marks for upward motion $\ddot{x}$].
$\Rightarrow v^2 = \frac{g}{k} \Rightarrow v = \sqrt{\frac{g}{k}}$ is Terminal velocity		
(ii) Upwards:		
$\uparrow mkv^2$ $M\ddot{x} = -mkv^2 - mg$		
$\downarrow mg$ $\ddot{x} = -kv^2 - g$		
$\ddot{x} = -(g + kv^2)$		
$v \frac{dv}{dx} = -(g + kv^2)$		✓ for correct mto $\ddot{x}$ or $\int$.
max ht: $\int_0^H \frac{v dv}{g + kv^2} = \int_0^H \frac{1}{g + kv^2} dx$		
initial $\rightarrow v$		
$H - 0 = -\frac{1}{2k} \int_0^H \frac{2k v dv}{g + kv^2} = -\frac{1}{2k} \int_0^H \frac{2k v dv}{g + kv^2}$		
$= -\frac{1}{2k} \left[\ln(g + kv^2) \right]_0^H$		✓ for correct integration

NSBHS Maths Ext. 2. Trial solution - 2004.		Marks
$H = -\frac{1}{2k} [\ln g - \ln(kv^2 + g)]$		
$= -\frac{1}{2k} [\ln(kv^2 + g) - \ln g]$		
$= -\frac{1}{2k} \ln \left(\frac{kv^2 + g}{g} \right)$		✓ for correct manipulation to get the given answer.
$= -\frac{1}{2k} \ln \left(\frac{kv^2}{g} + 1 \right)$ But $v^2 = \frac{g}{k}$		
$= -\frac{1}{2k} \ln \left(\frac{k \times \frac{g}{k}}{g} + 1 \right) = -\frac{1}{2k} \ln 2$		
(ii) Longer Approach: is to find the general x equation and then sub $v=0$ to get H:		or
2nd approach:		
(b) (i) $v \frac{dv}{dx} = -(g + kv^2)$		
$-\int \frac{v dv}{g + kv^2} = + \int \frac{1}{g + kv^2} dx$		✓ for correct $\ddot{x}$ or $\int$
$x = -\frac{1}{2k} \int \frac{2k v dv}{g + kv^2} = -\frac{1}{2k} \ln(g + kv^2)$		
$= -\frac{1}{2k} \ln \left(\frac{g + kv^2}{g} \right) = -\frac{1}{2k} \ln \left(\frac{g + kv^2}{g} \right)$		✓ for correct sign.
at max ht. $v=0$.		
$H = x = -\frac{1}{2k} \ln \left(\frac{g + kv^2}{g} \right)$ But $v^2 = \frac{g}{k}$		✓ for manipulation to get correct given answer
$H = -\frac{1}{2k} \ln \left(1 + \frac{g}{g} \times \frac{g}{k} \right) = -\frac{1}{2k} \ln 2$		
(ii) 3rd approach: Finding constants.		or
longer! (b) (i) $x = \int \frac{v dv}{g + kv^2}$		
$-x = \frac{1}{2k} \ln(g + kv^2) + C$		✓ for correct x .
when $x=0$, $v = \sqrt{\frac{g}{k}}$. $0 = \frac{1}{2k} \ln \left(g + \frac{g}{k} \right) + C$		
$C = -\frac{1}{2k} \ln 2$ or $C = -\frac{1}{2k} \ln(g + kv^2)$		
$+x = -\left[\frac{1}{2k} \ln(g + kv^2) \right] + \frac{1}{2k} \ln 2$		✓ for correct sign
$x = \frac{1}{2k} \ln \left(\frac{2}{g + kv^2} \right)$ or $x = \frac{1}{2k} \ln \left(\frac{g + kv^2}{g} \right)$		✓ for correct manipulation.
then showing when $x=H$, $v=0$ that		

NSBHS Ext 2 - Trial Solutions 2004	Marks.
(vi) Upward Mtn: $\ddot{x} = -(g + kv^2)$ $\frac{dv}{dt} = -(g + kv^2)$	
max ht. $\rightarrow \int_0^T dt = - \int_0^0 \frac{dv}{g + kv^2}$	✓ for correct $\ddot{x}$ or $\int$
initial $\rightarrow \int_0^0 \frac{dv}{g + kv^2} = - \int_0^0 \frac{dv}{g + kv^2}$	
$T = -\frac{1}{k} \int_0^0 \frac{dv}{\left(\frac{g}{k} + v^2\right)} = -\frac{1}{k} \frac{1}{\sqrt{\frac{g}{k}}} \left[\tan^{-1} \sqrt{\frac{k}{g}} v \right]_0^0$	✓ for correct indefinite $\int$ answer.
$T = -\frac{1}{k} \frac{1}{\sqrt{\frac{g}{k}}} \left[\tan^{-1} 0 - \tan^{-1} \sqrt{\frac{k}{g}} V \right]$	
But $-\frac{1}{k} \frac{1}{\sqrt{\frac{g}{k}}} = -\frac{1}{kV} ; \tan^{-1} \sqrt{\frac{k}{g}} V = \tan^{-1} \frac{1}{V} \times V$	✓ for correct T equation
$T = -\frac{1}{kV} \left[-\tan^{-1} 1 \right] = -\frac{1}{kV} \times \left(-\frac{\pi}{4} \right)$	✓ for manipulation to get correct answer.
But $\frac{1}{k} = \frac{V^2}{g} \Rightarrow T = \frac{V^2}{g} \times \frac{1}{V} \times \frac{\pi}{4} = \frac{V \pi}{4g}$	
2nd (Longer) Approach: get the time equation in general and then sub. in $v=0$ (max ht.)	OR
i.e. $\int_0^v \frac{dv}{g + kv^2} = - \int_0^t dt$	✓ for correct $\int$ or $\ddot{x}$
$v = 0 \rightarrow t = \frac{1}{k} \int_0^v \frac{dv}{\frac{g}{k} + v^2}$	
$t = \frac{1}{k} \frac{1}{\sqrt{\frac{g}{k}}} \left[\tan^{-1} \left(\sqrt{\frac{k}{g}} v \right) \right]_0^v$	✓ for correct integration answer (indefinite)
$t = -\frac{1}{k} \times \frac{1}{V} \left[\tan^{-1} \sqrt{\frac{k}{g}} v - \tan^{-1} \sqrt{\frac{k}{g}} \times V \right]$	
$= -\frac{1}{kV} \left[\tan^{-1} \sqrt{\frac{k}{g}} v - \tan^{-1} \frac{1}{V} \times V \right]$	✓ for correct definite $\int$ answer
at max ht., $v=0$	
$T = -\frac{1}{kV} \left[-\tan^{-1} 1 \right] = \frac{1}{kV} \times \frac{\pi}{4}$ which could be shown as above	✓ for manipulation.
$T = \frac{V^2}{g} \times \frac{1}{V} \times \frac{\pi}{4} = \frac{V \pi}{4g}$	

NSBHS Ext 2 - Trial Solutions 2004	Marks.
(vi) Downward Motion: $\ddot{x} = \frac{dv}{dt} = g - kv^2$	
$\int_0^v \frac{dv}{g - kv^2} = \int_0^t dt$	✓ for correct expression
$t = \int_0^v \frac{dv}{\left(\frac{g}{k} - v^2\right)} = \int_0^v \frac{dv}{\left(\frac{g}{k} - v^2\right)}$	
$\frac{1}{g - kv^2} = \frac{a}{\sqrt{\frac{g}{k}} - v} + \frac{b}{\sqrt{\frac{g}{k}} + v}$	
$a = \lim_{v \rightarrow \sqrt{\frac{g}{k}}} \frac{1}{\sqrt{\frac{g}{k}} + v} = \frac{1}{\sqrt{\frac{g}{k}} + \sqrt{\frac{g}{k}}} = \frac{1}{2\sqrt{\frac{g}{k}}}$	
$b = \lim_{v \rightarrow -\sqrt{\frac{g}{k}}} \frac{1}{\sqrt{\frac{g}{k}} - v} = \frac{1}{\sqrt{\frac{g}{k}} - (-\sqrt{\frac{g}{k}})} = \frac{1}{2\sqrt{\frac{g}{k}}}$	✓ for correct resolution of partial fractions.
$\therefore t = \frac{1}{2\sqrt{\frac{g}{k}}} \left[\int_0^v \frac{dv}{\sqrt{\frac{g}{k}} - v} + \int_0^v \frac{dv}{\sqrt{\frac{g}{k}} + v} \right]$	
$= \frac{1}{2\sqrt{\frac{g}{k}}} \left[-\ln \left(\sqrt{\frac{g}{k}} - v \right) + \ln \left(\sqrt{\frac{g}{k}} + v \right) \right]$	
$= \frac{1}{2\sqrt{\frac{g}{k}}} \times \frac{1}{\sqrt{\frac{g}{k}}} \ln \left(\frac{\sqrt{\frac{g}{k}} + v}{\sqrt{\frac{g}{k}} - v} \right)$	✓ for integration expansion
$= \frac{1}{2\sqrt{\frac{g}{k}}} \ln \left(\frac{\frac{\sqrt{\frac{g}{k}}}{\sqrt{\frac{g}{k}}} + v}{\frac{\sqrt{\frac{g}{k}}}{\sqrt{\frac{g}{k}}} - v} \right) = \frac{1}{2\sqrt{\frac{g}{k}}} \ln \left(\frac{1 + v}{1 - v} \right)$	✓ for manipulation to get given answer
2nd approach: (Longer).	OR
$\frac{dv}{dt} = g - kv^2 = k \left(\frac{g}{k} - v^2 \right) = k(V^2 - v^2)$	
$\therefore \int_0^v \frac{dv}{V^2 - v^2} = \int_0^t \frac{g}{V^2} dt$	✓ for correct expression
$\frac{1}{V^2 - v^2} = \frac{1}{2V} \left[\frac{1}{V - v} \right] + \frac{1}{2V} \left[\frac{1}{V + v} \right]$ partial fraction.	✓ for correct partial fractions.
$\therefore \frac{g}{V^2} = \int_0^v \frac{dv}{V^2 - v^2} = \frac{1}{2V} \left[-\ln(V - v) + \ln(V + v) \right]_0^v$	✓ for correct
$\frac{g}{V^2} = \frac{1}{2V} \left[\ln \left(\frac{V + v}{V - v} \right) \right]_0^v = \frac{1}{2V} \left[\ln \left(\frac{V + v}{V - v} \right) - \ln 1 \right]$	
$\frac{g}{V^2} = \frac{1}{2V} \ln \left(\frac{V + v}{V - v} \right)$ But $V = \sqrt{\frac{g}{k}}$	✓ for correct manipulation.
$t = \frac{V}{2g} \ln \left(\frac{V + v}{V - v} \right) = \frac{1}{2g} \sqrt{\frac{g}{k}} \ln \left(\frac{V + v}{V - v} \right) = 1$	
$t = \frac{1}{2g} \ln \left(\frac{V + v}{V - v} \right)$	

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Question 8:	
(A) $I_n = \int_0^1 (1-x^2)^n dx, n \geq 0$	
$u = (1-x^2)^n \quad dv = dx$	
$du = n(1-x^2)^{n-1}(-2x) \quad v = x$	
$I_n = \int_0^1 u dv = \int_0^1 x du$	
$= \left[x(1-x^2)^n \right]_0^1 - \int_0^1 x(-2x)(1-x^2)^{n-1} dx$	✓ for correct start of integration of parts
$= 0 + 2n \int_0^1 x^2(1-x^2)^{n-1} dx$	
$= -2n \int_0^1 x^2(1-x^2)^{n-1} dx$	
$= -2n \int_0^1 (1-x^2)^{n-1} (1-x^2) dx$	✓ for answer
$= -2n \left[\int_0^1 (1-x^2)^{n-1} dx - \int_0^1 (1-x^2)^n dx \right]$	✓ for correct manipulation to get the relation
$= -2n [I_n - I_{n+1}]$	
$I_n = -2n I_n + 2n I_{n+1}$	
$I_n(1+2n) = 2n I_{n+1}$	
$I_n = \frac{2n I_{n+1}}{1+2n} \text{ for } n \geq 1$	
(ii) $I_0 = \int_0^1 (1-x^2)^0 dx = \int_0^1 dx = [x]_0^1 = 1$	
$I_n = \frac{2n}{1+2n} \left[\frac{I_{n-1}}{1+2(n-1)} \left[\frac{I_{n-2}}{1+2(n-2)} \left[\frac{I_{n-3}}{1+2(n-3)} \dots \left[\frac{I_2}{1+2(2)} \right] \right] \right] \right] \times \frac{I_1}{1+2(1)}$	✓ for correct resolution to I_0
$\left(\frac{2 \times 2}{5} \right) \times \frac{2}{3} \times 1$	
$I_n = \frac{2^n [n(n-1)(n-2) \dots 3 \times 2 \times 1]}{(1+2n)(2n-1)(2n-3)(2n-5) \dots 7 \times 5 \times 3 \times 1}$	
$= \frac{2^n n!}{(2n+1)(2n-1)(2n-3)(2n-5) \dots 7 \times 5 \times 3 \times 1}$	✓ for relation
$\downarrow$ odd Numbers	
For the smarties odd numbers = $\frac{(2n+1)!}{2^n n!}$	
$I_n = \frac{2^n n!}{\frac{(2n+1)!}{2^n n!}} = \frac{2^{2n} (n!)^2}{(2n+1)!}$	

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Q8 (b)	
Model Answer:	Award 1 mark for each of the reasons:
Let $\angle DAB = x$	
$\therefore \angle DCE = \angle DAB = x$ (Exterior $\angle$ of cyclic quad. = opp. interior $\angle$)	✓
$\therefore \angle DFE = \angle DCE = x$ ($\angle$ s in same segment arc)	✓
$\therefore \angle DFG = 180^\circ - x$ (GFE is a st. line)	✓
$\therefore \angle DAG + \angle DFG = x + (180 - x) = 180^\circ$	
[eg: $\angle DFE = x = \angle DAG$ (Exterior $\angle$ theorem...)]	
$\therefore GFDA$ is a cyclic quadrilateral	✓
since a pair of its opp. $\angle$ s are supplementary	
(c) $u_1 = 1; u_{n+1} = \frac{1}{2} \left[\frac{u_n + 2}{u_n} \right]$ when $n \geq 1; u_n = \frac{(1-\sqrt{2})^n}{1+\sqrt{2}}$	
step 1: Prove true for $n=1$ (S(1))	
$u_1 = 1; u_2 = \frac{1+\sqrt{2}}{1+\sqrt{2}} = 1+\sqrt{2}$	
RHS = $\frac{(1-\sqrt{2})^2}{(1+\sqrt{2})} = \frac{(1-\sqrt{2})}{1+\sqrt{2}} = 1-\sqrt{2}$	✓ for 1 st step
step 2: If $S(k)$ is true: $\frac{u_k - \sqrt{2}}{u_k + \sqrt{2}} = \left(\frac{1-\sqrt{2}}{1+\sqrt{2}} \right)^{k-1}$ then	
now prove $S(k+1) = \frac{u_{k+1} - \sqrt{2}}{u_{k+1} + \sqrt{2}} = \left(\frac{1-\sqrt{2}}{1+\sqrt{2}} \right)^k$	
Proof: $\frac{u_{k+1} - \sqrt{2}}{u_{k+1} + \sqrt{2}} = \frac{\frac{1}{2} \left[\frac{u_k + 2}{u_k} \right] - \sqrt{2}}{\frac{1}{2} \left[\frac{u_k + 2}{u_k} \right] + \sqrt{2}} = \frac{\frac{1}{2} \left[\frac{u_k + 2 - 2\sqrt{2}u_k}{u_k} \right]}{\frac{1}{2} \left[\frac{u_k + 2 + 2\sqrt{2}u_k}{u_k} \right]}$	✓

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$\therefore \frac{U_{k+1} - \sqrt{2}}{U_{k+1} + \sqrt{2}} = \frac{U_k^2 + 2 - 2\sqrt{2} U_k}{U_k^2 + 2\sqrt{2} U_k + 2}$ $= \frac{(U_k - \sqrt{2})^2}{(U_k + \sqrt{2})^2}$ $= \left(\frac{U_k - \sqrt{2}}{U_k + \sqrt{2}} \right)^2$ from assumption $\rightarrow \left[\left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^{2^{k-1}} \right]^2$ $= \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^{2^{k-1} \cdot 2}$ $= \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^{2^k}$	✓ ✓
$\therefore S(k+1)$ is true whenever $S(k)$ is true. $\Rightarrow S(k)$ is true. (ii) $-1 < \frac{1 - \sqrt{2}}{1 + \sqrt{2}} < 1$ let $r = \frac{1 - \sqrt{2}}{1 + \sqrt{2}}$ As $n \rightarrow \infty$: ratio $= \left(\frac{1 - \sqrt{2}}{1 + \sqrt{2}} \right)^{2^{n-1}} \rightarrow 0$ since $r < 1$, $r^n \rightarrow 0$.	✓ for justification of why RHS $\rightarrow 0$
$\therefore \frac{U_n - \sqrt{2}}{U_n + \sqrt{2}} \rightarrow 0$ $\therefore U_n - \sqrt{2} \rightarrow 0$ $\therefore U_n \rightarrow \sqrt{2}$	✓ for showing the answer $U_n \rightarrow \sqrt{2}$.
Thus for n sufficiently large, $U_n \rightarrow 2$.	